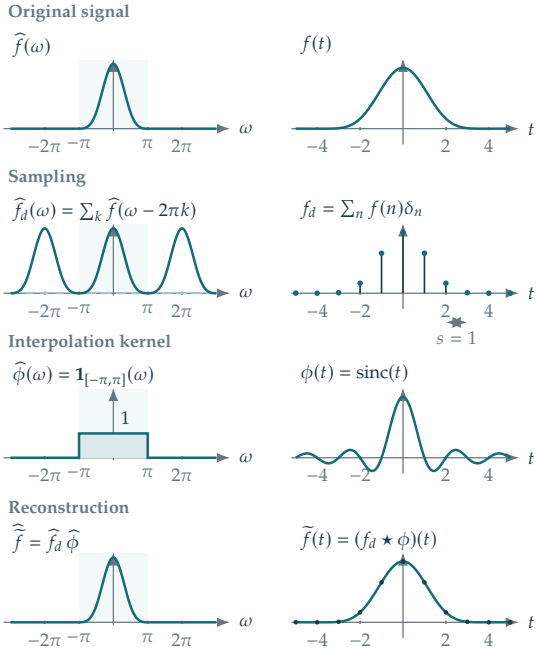
Shannon condition holds

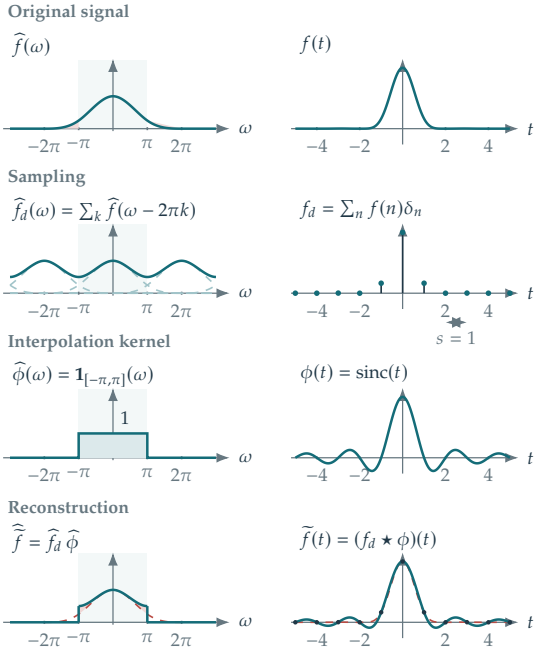
$f(t) = \operatorname{sinc}^4(t/4)$



$\widetilde{f} = f$: exact reconstruction

Shannon condition is violated

$f(t) = \operatorname{sinc}^4(t/2)$



$\widetilde{f} \neq f$, although $\widetilde{f}(n) = f(n)$

$$\operatorname{sinc}(t) = \frac{\sin(\pi t)}{\pi t}, \qquad \widetilde{f}(t) = \sum_{n \in \mathbb{Z}} f(n) \operatorname{sinc}(t - n)$$

Pale band: $[-\pi, \pi]$. Dashed spectral curves: replicas. Red shading: aliased mass.
Dashed red in the reconstruction row: the original signal and spectrum. Dots: the shared samples.